

Period-Three Trace Residues and a Minimal Separator on an Exceptional Quartic Hénon Fiber

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Abstract

Let $f_{m,a}(x, y) = (y + (x^m - a)^2, x)$ over an algebraically closed field of characteristic zero. We study trace moments on its nonreduced formal periodic schemes, rather than replacing those schemes by their reduced supports. The period-one multiplication spectrum is $0^{\times 2m}$, the formal exact-period-two spectrum is $2^{\times ((2m)^2 - 2m)}$, and normalized conjugacy is governed by the coordinate a^{2m-1} . For the period-three cyclic complete intersection, we prove the two-term identity

$$S_m(a, \varepsilon) = C_m \varepsilon^{3m} + D_m a^{2m-1} \varepsilon^{2m},$$

with $C_m = 0$ for odd m , and derive a finite nested-binomial certificate for D_m from a terminating normal-form recurrence and an exact Laurent fiber sum. In degree four, every normalized monic-centered map with formal fixed-point trace multiset 0^4 belongs to the complete fiber $p(x) = (x^2 - L)^2$. On this fiber the pointwise formal exact-period-three second trace moment is

$$-1296000 - 1572864L^3,$$

the exact-period-three zero-cycle has length 60, and the cyclewise moment is $-432000 - 524288L^3$. Since L^3 is precisely the normalized conjugacy coordinate while periods one and two are constant, period three is the minimal separator on this fiber. Universal nonvanishing of D_m remains open.

1 Introduction

Finite collections of periodic multipliers can determine a polynomial automorphism only after one understands their exceptional fibers. In the normalized Jacobian-minus-one Hénon category, Cantat and Dujardin identified the quartic family

$$(x, y) \mapsto (y + (x^2 - \lambda^2)^2, x)$$

whose period-one and period-two trace data are constant [1]. Thus neither the family nor its low-period blindness is new. The concrete question left by this obstruction is whether a later period supplies an effective coordinate on the whole exceptional fiber.

This note answers that question at period three for the complete normalized quartic fiber whose formal fixed-point trace multiset is 0^4 . Writing $L = \lambda^2$, we show that the pointwise formal exact-period-three second trace moment equals

$$-384(3375 + 4096L^3).$$

The coefficient of L^3 is nonzero, and L^3 is exactly the residual normalized conjugacy coordinate. Since the two lower formal trace multisets are constant, this gives a sharp minimal separator on that fiber. It does not give a global period cutoff for all quartic Hénon maps.

The quartic identity is the endpoint of a uniform algebraic mechanism. For every $m \geq 2$, three cyclic orbit equations form a monic complete intersection of rank $(2m)^3$. Their Jacobian determinant is also the trace of the third derivative iterate. Classical quotient-trace and global residue theory then converts the desired trace moment into a top normal-form coefficient [2]. Weighted homogeneity and a local degeneration leave only two parameter monomials. The remaining slope is evaluated by an explicit recurrence-to-Laurent-to-binomial argument. The conclusion is a formula for D_m , not a proof that $D_m \neq 0$ in every degree.

Our contributions are therefore the following.

- (i) We retain the nonreduced formal periodic algebras and compute the period-one and formal exact-period-two trace spectra, as well as the normalized quotient coordinate a^{2m-1} .
- (ii) We prove the uniform period-three law $S_m = C_m \varepsilon^{3m} + D_m a^{2m-1} \varepsilon^{2m}$, the odd- m identity $C_m = 0$, and a transparent finite certificate for D_m .
- (iii) We classify the complete normalized quartic fiber with formal fixed-point trace multiset 0^4 , compute its period-three affine moment by two source-level derivations, and prove its scoped minimality after a separate local-multiplicity argument.

The proof has two distinctions that cannot be suppressed. First, a trace element can be nonzero and nilpotent on a nonreduced scheme even when its formal multiplication spectrum contains only zero. Second, vanishing of the fixed *moment* does not determine the fixed scheme's local *multiplicity*. The former is handled in the finite quotient algebras; the latter is proved by formal transverse elimination before any exact-period length is subtracted.

2 Prior work and claim boundary

Normal forms and multiplier rigidity. Friedland–Milnor theory supplies the structural and conjugacy framework for plane polynomial automorphisms and generalized Hénon normal forms [5]. Cantat–Dujardin place this framework in a modern multiplier-rigidity setting and provide the direct quartic period-one/two obstruction used here [1]. Our equivalence $f_{m,a} \sim f_{m,b}$ if and only if $a^{2m-1} = b^{2m-1}$ is a proved specialization within the normalized one-factor category, not a new general normal-form theorem.

Residues and exact Hénon identities. Multidimensional residues, quotient-algebra traces, deformations to initial forms, and coefficient extraction are established tools [2]. Exact periodic-orbit sum rules for Hénon dynamics also predate this work [3]. Those contour and Fredholm-type sums use different weights from the raw finite-algebra trace power considered below. The present contribution is the family-specific complete intersection, the two-term reduction, its coefficient certificate, and the quartic separator; it is not the residue formalism itself.

Low periods and adjacent multiplier invariants. Low-period equations and stability for generalized Hénon maps have been studied explicitly, for example in the cubic area-preserving setting [4]. Small-cycle multiplier maps in one-dimensional polynomial moduli provide a close comparison but do not cover Jacobian-minus-one plane automorphisms [7]. Formal dynatomic cycles and higher-dimensional projective multiplier invariants supply adjacent language and background [8, 9]. Index-derived fixed-point relations in projective and polynomial-automorphism settings are further methodological neighbors [6, 10]. None of these sources is used as a direct proof of the identities below.

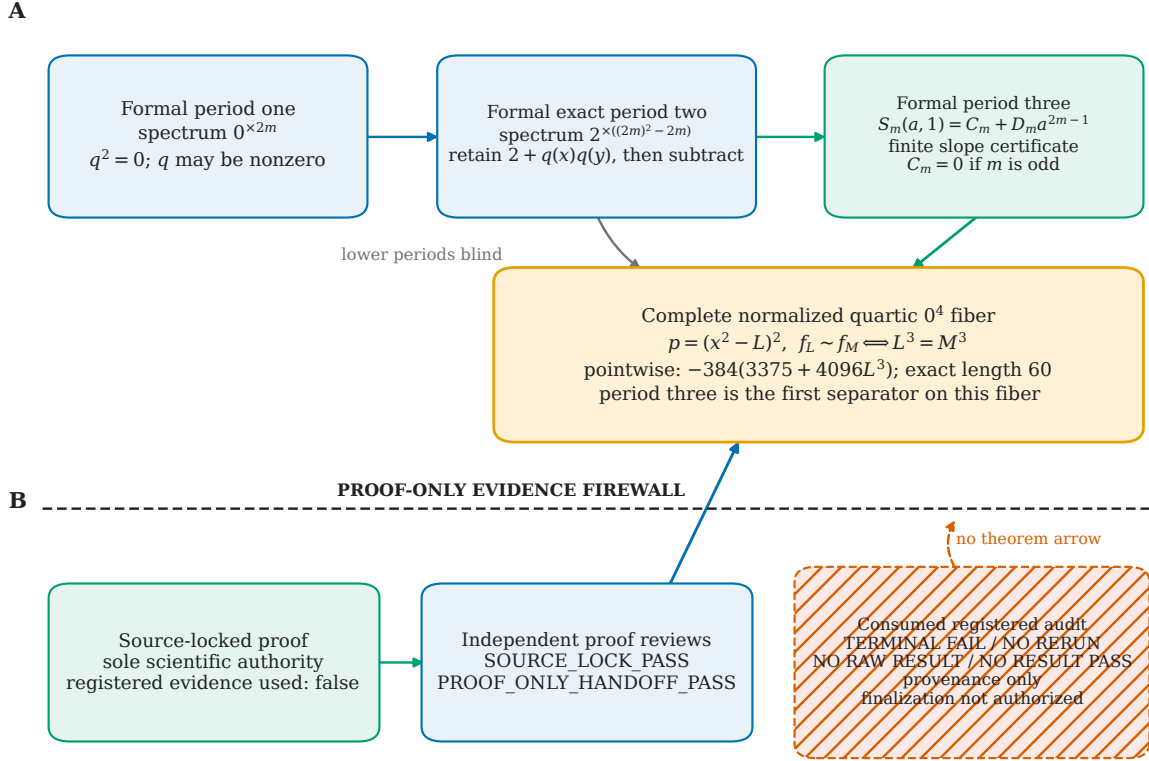


Figure 1: Theorem architecture and evidence boundary. The formal period-one and exact-period-two multiplication spectra are constant on the normalized family, with nilpotent corrections retained before passing to spectra. The period-three cyclic complete intersection yields the two-term moment law and, on the complete normalized quartic fiber with formal fixed-point trace multiset 0^4 , the exact pointwise moment $-384(3375 + 4096L^3)$, formal exact-period-three length 60, and minimal separation by L^3 . Every arrow into a theorem box originates in the source-locked proof and independent proof reviews. The consumed registered audit contributes only the dashed provenance branch: it terminally failed, supplies no scientific evidence, and authorizes neither a result claim nor finalization.

Bounded novelty statement. The literature search underlying this note located no prior explicit formula for the period-three moment, the all- m coefficient certificate, or the scoped quartic minimal-separator theorem. This is a bounded absence statement, not a historical-priority proof. The claim-safe delta is limited to the source-proved period-three results. In particular, we do not claim discovery of the exceptional quartic family, novelty of formal cycles or global residues, or a first exact Hénon period-three calculation of any kind.

3 Formal periodic algebras and main results

Let $\mathbb{k}$ be an algebraically closed field of characteristic zero. For an integer $m \geq 2$ and $a \in \mathbb{k}$, set

$$p_{m,a}(x) = (x^m - a)^2, \quad q_{m,a}(x) = p'_{m,a}(x) = 2mx^{m-1}(x^m - a), \quad (1)$$

and

$$f_{m,a}(x, y) = (y + p_{m,a}(x), x). \quad (2)$$

The Jacobian determinant of $f_{m,a}$ is -1 .

Definition 3.1 (Formal multiset). Let B be a finite-dimensional $\mathbb{k}$ -algebra and $h \in B$. The formal multiset of values of h is the multiset of roots of $\det(TI - M_h)$, counted with algebra dimension. Here M_h is multiplication by h . This convention retains nilpotent scheme length while attaching to each support point the residue-field value of h .

For period two, “formal exact period two” means the formal period-two spectral zero-cycle minus the embedded formal fixed spectral zero-cycle. For prime period three, we use the analogous subtraction, but we will prove separately that the local fixed length inside $\text{Fix}(f^3)$ is the expected one.

Write

$$\nu = 2m - 1, \quad X_{\text{top}} = x_0^\nu x_1^\nu x_2^\nu. \quad (3)$$

All cyclic indices below lie in $\mathbb{Z}/3\mathbb{Z}$.

Theorem 3.2 (Uniform formal period-three law). *For the family (2):*

(a) *the formal period-one trace multiset is $0^{\times 2m}$, and the formal exact-period-two trace multiset is $2^{\times ((2m)^2 - 2m)}$;*

(b) *within normalized monic-centered Hénon moduli,*

$$f_{m,a} \sim f_{m,b} \iff a^{2m-1} = b^{2m-1};$$

(c) *the formal period-three m -th trace moment satisfies*

$$S_m(a, \varepsilon) = C_m \varepsilon^{3m} + D_m a^{2m-1} \varepsilon^{2m} \quad (4)$$

for integers C_m, D_m , and $C_m = 0$ whenever m is odd;

(d) *defining*

$$H(r, k) = \sum_{\substack{u+v=k \\ 2u \leq r, 2v \leq r}} \binom{k}{u} \binom{r}{2u} \binom{r}{2v}, \quad (5)$$

$$A_{m,r} = \sum_{k=\lceil m/2 \rceil}^{\min(m-1, r)} (-1)^{r+k} \binom{m-1}{2(m-k)-1} H(r, k), \quad (6)$$

with an empty sum equal to zero, the slope has the exact certificate

$$D_m = 3 \sum_{j=0}^{\lfloor m/2 \rfloor} \binom{m+1}{j} (2m)^{3m+3-2j} A_{m,m-j}. \quad (7)$$

No assertion that $D_m \neq 0$ for all m is included.

Theorem 3.3 (Complete quartic fiber and minimal separator). *Let p be any monic-centered quartic, and set $f_p(x, y) = (y + p(x), x)$. Its formal fixed-point trace multiset is 0^4 if and only if*

$$p(x) = (x^2 - L)^2$$

for a unique $L \in \mathbb{k}$. On this complete normalized fiber,

$$S_2^{(3)}(L) = -1296000 - 1572864L^3 = -384(3375 + 4096L^3). \quad (8)$$

Moreover,

$$f_{2,L} \sim f_{2,M} \iff L^3 = M^3.$$

The period-one and formal exact-period-two trace multisets are constant on the fiber, whereas (8) is affine with nonzero slope in L^3 . Hence period three is the minimal separating period on this fiber. The formal exact-period-three zero-cycle has pointwise length 60, and its cyclewise second trace moment is

$$-432000 - 524288L^3.$$

Scope warning 3.4. The minimality conclusion in theorem 3.3 concerns only the complete normalized quartic fiber with formal fixed-point trace multiset 0^4 . It is neither a theorem for all quartic Hénon maps nor a global equality for a degree-four period cutoff.

4 Low periods and normalized conjugacy

Write $p = p_{m,a}$ and $q = q_{m,a}$. A fixed point satisfies $x = y$ and $p(x) = 0$, so its coordinate algebra is

$$B_1 = \mathbb{k}[x]/(p) = \mathbb{k}[x]/((x^m - a)^2), \quad \dim_{\mathbb{k}} B_1 = 2m. \quad (9)$$

Within B_1 ,

$$q = 2mx^{m-1}(x^m - a), \quad q^2 = 0.$$

Since

$$Df_{m,a}(x, y) = \begin{pmatrix} q(x) & 1 \\ 1 & 0 \end{pmatrix},$$

the trace element on the fixed algebra is q . Multiplication by q is nilpotent, hence its characteristic polynomial is T^{2m} . This gives the formal multiset $0^{\times 2m}$. The statement is spectral: when $a \neq 0$, the class of q is generally not zero in the nonreduced algebra.

A direct iteration gives

$$f_{m,a}^2(x, y) = (x + p(y + p(x)), y + p(x)).$$

The fixed equations for f^2 reduce to $p(x) = p(y) = 0$, so

$$B_2 = \mathbb{k}[x, y]/(p(x), p(y)), \quad \dim_{\mathbb{k}} B_2 = (2m)^2. \quad (10)$$

The derivative trace is

$$\mathrm{Tr}(Df_{m,a}^2) = 2 + q(x)q(y).$$

Because $(q(x)q(y))^2 = 0$, its sole formal eigenvalue is 2, with multiplicity $(2m)^2$. On the embedded fixed algebra the same trace is $2 + q^2 = 2$. Subtracting the length- $2m$ fixed spectral cycle yields $2 \times ((2m)^2 - 2m)$. Nothing in this argument reduces either scheme, so it remains valid at $a = 0$.

We next identify the normalized moduli coordinate. For $h_c(x, y) = (cx, cy)$,

$$\begin{aligned} h_c^{-1} f_{m,a} h_c(x, y) &= \left(y + c^{-1}(c^m x^m - a)^2, x \right) \\ &= \left(y + c^{2m-1}(x^m - ac^{-m})^2, x \right). \end{aligned} \quad (11)$$

The image stays monic exactly when $c^{2m-1} = 1$, and then the new parameter is $b = ac^{-m}$. Thus conjugacy implies $b^{2m-1} = a^{2m-1}$.

Conversely, if $a^{2m-1} = b^{2m-1}$ and $a \neq 0$, then $\zeta = b/a$ is a $(2m-1)$ -st root of unity. Since $\gcd(m, 2m-1) = 1$, exponentiation by $-m$ is an automorphism of this root group; choose c with $c^{2m-1} = 1$ and $c^{-m} = \zeta$. Equation (11) gives the desired conjugacy. The case $a = 0$ forces $b = 0$.

For necessity within the normalized category, normal-form uniqueness reduces a polynomial conjugacy to a simultaneous affine change $(x, y) \mapsto (cx + d, cy + d)$. The coefficient of x^{2m-1} in the transformed degree- $2m$ polynomial is $2md/c$. Centering and characteristic zero force $d = 0$, so the diagonal calculation exhausts the normalized conjugacies.

5 The cyclic complete intersection and trace residue

Represent a length-three orbit by states (x_i, x_{i-1}) . Introduce the deformation

$$F_i = (x_i^m - a)^2 + \varepsilon(x_{i-1} - x_{i+1}), \quad q_i = 2mx_i^{m-1}(x_i^m - a). \quad (12)$$

At $\varepsilon = 1$, the equations $F_i = 0$ are exactly the cyclic equations for $\text{Fix}(f_{m,a}^3)$. Their Jacobian matrix is

$$\left(\frac{\partial F_i}{\partial x_j} \right) = \begin{pmatrix} q_0 & -\varepsilon & \varepsilon \\ \varepsilon & q_1 & -\varepsilon \\ -\varepsilon & \varepsilon & q_2 \end{pmatrix},$$

and its determinant is

$$t_\varepsilon = q_0 q_1 q_2 + \varepsilon^2(q_0 + q_1 + q_2). \quad (13)$$

If $M(q) = \begin{pmatrix} q & 1 \\ 1 & 0 \end{pmatrix}$, then $Df^3 = M(q_2)M(q_1)M(q_0)$. Multiplication of these three matrices shows that $\text{Tr}(Df^3) = q_0 q_1 q_2 + q_0 + q_1 + q_2 = t_1$.

Let

$$\mathcal{A}_m = \mathbb{k}[a, \varepsilon, x_0, x_1, x_2] / (F_0, F_1, F_2). \quad (14)$$

For any degree-compatible monomial order, the leading monomials are $x_0^{2m}, x_1^{2m}, x_2^{2m}$. They are pairwise coprime, so the three relations form a Gröbner basis over $\mathbb{k}[a, \varepsilon]$. Consequently

$$x_0^{e_0} x_1^{e_1} x_2^{e_2}, \quad 0 \leq e_i < 2m,$$

form a free basis of $\mathcal{A}_m$, of rank $(2m)^3$. The leading homogeneous forms have no common point in $\mathbb{P}^2$, hence no intersection length is lost at infinity.

Define

$$S_m(a, \varepsilon) = \text{Tr}_{\mathcal{A}_m / \mathbb{k}[a, \varepsilon]}(M_{t_\varepsilon}^m). \quad (15)$$

The Jacobian of the complete intersection is t_ε . The standard trace-residue identity, valid also for nonreduced complete intersections, gives

$$S_m(a, \varepsilon) = \text{Res}_{F_0, F_1, F_2}(t_\varepsilon^{m+1}). \quad (16)$$

For these monic leading forms, the residue is the top normal-form coefficient:

$$\text{Res}(G) = \left[x_0^{2m-1} x_1^{2m-1} x_2^{2m-1} \right] \text{NF}(G). \quad (17)$$

In particular, $S_m \in \mathbb{Z}[a, \varepsilon]$.

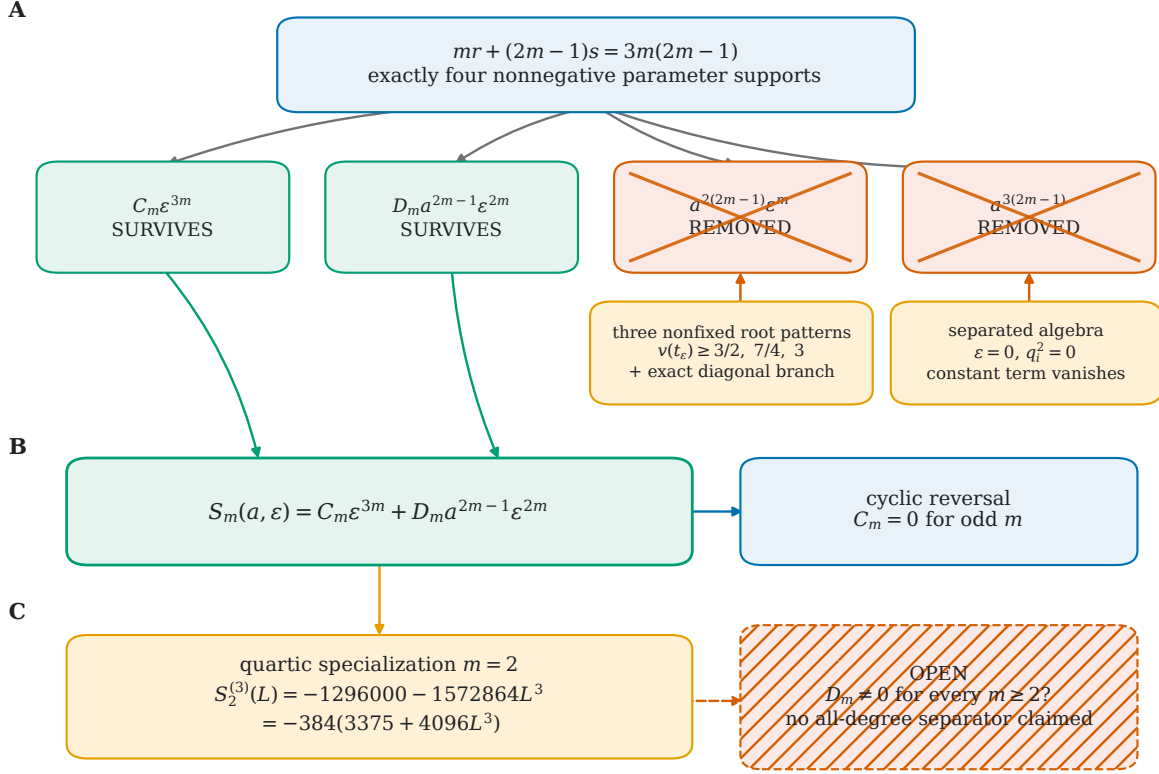


Figure 2: Weighted support and the two-term law. Weighted homogeneity permits exactly four parameter monomials in $S_m(a, \varepsilon)$. The separated algebra at $\varepsilon = 0$ removes $a^{3(2m-1)}$, and the three nonfixed degeneration patterns together with the exact diagonal branch remove $a^{2(2m-1)}\varepsilon^m$. The surviving terms are $C_m \varepsilon^{3m}$ and $D_m a^{2m-1} \varepsilon^{2m}$; cyclic reversal additionally gives $C_m = 0$ for odd m . For $m = 2$, source-level coefficient derivations give $S_2^{(3)}(L) = -1296000 - 1572864L^3$. The diagram makes no universal separation claim: $D_m \neq 0$ for every $m \geq 2$ remains open.

6 The two-term law

Assign weights

$$\text{wt}(x_i) = 1, \quad \text{wt}(a) = m, \quad \text{wt}(\varepsilon) = 2m - 1 = \nu.$$

Each F_i has weight $2m$, every q_i has weight ν , and t_ε has weight 3ν . Therefore each monomial $a^r \varepsilon^s$ in S_m satisfies

$$mr + \nu s = 3m\nu.$$

Since $\gcd(m, \nu) = 1$, the only nonnegative solutions are

$$(r, s) = (0, 3m), (\nu, 2m), (2\nu, m), (3\nu, 0).$$

Thus

$$S_m = c_{0,m} \varepsilon^{3m} + c_{1,m} a^\nu \varepsilon^{2m} + c_{2,m} a^{2\nu} \varepsilon^m + c_{3,m} a^{3\nu}. \quad (18)$$

At $\varepsilon = 0$, the algebra is the tensor product of three copies of $\mathbb{k}[a, x]/((x^m - a)^2)$. In each factor $q_i^2 = 0$, whereas $t_0 = q_0 q_1 q_2$. Because $m \geq 2$, we have $t_0^m = 0$, and hence $S_m(a, 0) = 0$. Therefore $c_{3,m} = 0$.

It remains to remove the term of ε -order m . Fix $a \neq 0$ and work over an algebraic closure of the Puiseux field in ε , normalized by $v(\varepsilon) = 1$. Flatness clusters all $(2m)^3$ solutions at triples $(\alpha_0, \alpha_1, \alpha_2)$ of roots of $x^m = a$. Put $x_i = \alpha_i + \delta_i$. Since the roots are nonzero and simple before squaring,

$$(x_i^m - a)^2 = c_i \delta_i^2 + O(\delta_i^3), \quad q_i = d_i \delta_i + O(\delta_i^2),$$

with $c_i, d_i \neq 0$. Three root patterns exhaust the possibilities.

All roots distinct. Every constant coupling $\alpha_{i-1} - \alpha_{i+1}$ is nonzero. Balancing its valuation 1 against $2v(\delta_i)$ gives $v(q_i) = 1/2$, hence

$$v(t_\varepsilon) \geq \frac{3}{2}. \quad (19)$$

Exactly two roots equal. After cyclic relabeling take $\alpha_0 = \alpha_1 \neq \alpha_2$. The first two equations give $v(q_0) = v(q_1) = 1/2$. In the third equation the coupling has valuation at least $3/2$. If $v(\delta_2) < 3/4$, its square would be the unique least-valuation term. Therefore

$$v(q_2) \geq \frac{3}{4}, \quad v(t_\varepsilon) \geq \frac{7}{4}. \quad (20)$$

All roots equal. Let r_0 be the minimum valuation of a nonzero δ_i . Coupling terms have valuation at least $1 + r_0$. If $r_0 < 1$, a coordinate with valuation r_0 would contribute an uncanceled square of valuation $2r_0 < 1 + r_0$. Hence every nontrivial branch has $r_0 \geq 1$, and

$$v(t_\varepsilon) \geq 3. \quad (21)$$

The exact diagonal fixed branch is handled separately in lemma 9.1; its moment contribution vanishes identically.

The trace of multiplication by an element of a finite algebra is the sum of its support values weighted by local length; nilpotent parts have trace zero. The three lower bounds show that every nonfixed cluster contributes to S_m with ε -valuation strictly greater than m : the bounds are $3m/2$, $7m/4$, and $3m$, respectively. Summation cannot decrease valuation. In (18), the only surviving candidate of exact order m is $c_{2,m} a^{2\nu} \varepsilon^m$. Since $a \neq 0$, it follows that $c_{2,m} = 0$. Setting $C_m = c_{0,m}$ and $D_m = c_{1,m}$ proves (4).

Finally, the coordinate reversal $x_i \mapsto x_{-i}$ sends the system with ε to the system with $-\varepsilon$, while t_ε is unchanged. Therefore $S_m(a, \varepsilon) = S_m(a, -\varepsilon)$. The D_m -term is always even in ε ; if m is odd, $3m$ is odd, forcing $C_m = 0$.

7 A transparent coefficient certificate for D_m

This section gives the full coefficient extraction. It is an algebraic proof for symbolic m , not an interpolation from finitely many degrees.

For every integer exponent $e_i \geq 2m$, the relation $F_i = 0$ yields

$$x_i^{e_i} \equiv 2ax_i^{e_i-m} - a^2x_i^{e_i-2m} - \varepsilon x_i^{e_i-2m}x_{i-1} + \varepsilon x_i^{e_i-2m}x_{i+1} \pmod{(F_0, F_1, F_2)}. \quad (R)$$

For $\mathbf{e} = (e_0, e_1, e_2) \in \mathbb{Z}^3$ and $\alpha, \beta \in \mathbb{Z}$, define

$$\mathcal{R}_m(\mathbf{e}; \alpha, \beta) = \left[a^\alpha \varepsilon^\beta X_{\text{top}} \right] \text{NF}(x_0^{e_0} x_1^{e_1} x_2^{e_2}). \quad (22)$$

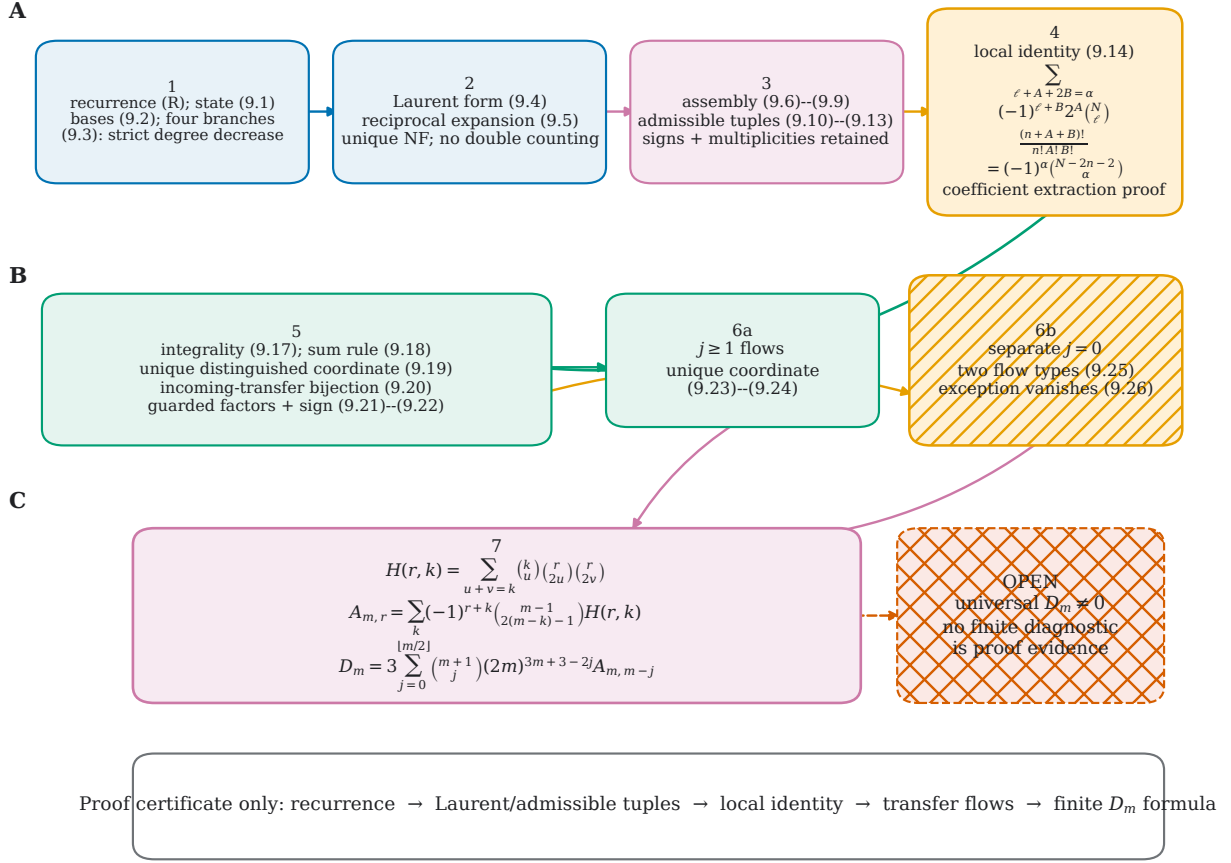


Figure 3: Transparent coefficient-certificate pipeline. The four-branch recurrence and base cases terminate by strict total-degree decrease; the Laurent coefficient makes the reduction order independent and groups branch interleavings without double counting. Admissible tuples retain every sign, multiplicity, Laurent exponent, and parameter degree. The local identity (9.14), proved by one-variable coefficient extraction, collapses each signed fiber. For $j \geq 1$, a unique distinguished coordinate and an incoming-transfer bijection give the guarded H/A factors; $j = 0$ is handled by a separate exhaustive flow classification, including the vanishing exceptional pattern. Assembly yields the finite formula for D_m . Universal $D_m \neq 0$ is explicitly open, and no runtime or finite diagnostic is part of this proof.

It is zero if some $e_i < 0$, or if $\alpha < 0$ or $\beta < 0$. On the standard box its complete base cases are

$$\mathcal{R}_m(\mathbf{e}; \alpha, \beta) = \begin{cases} 1, & \mathbf{e} = (\nu, \nu, \nu), (\alpha, \beta) = (0, 0), \\ 0, & 0 \leq e_i < 2m \text{ for all } i, \text{ otherwise.} \end{cases} \quad (23)$$

If $e_i \geq 2m$, (R) gives the four signed branches

$$\begin{aligned} \mathcal{R}_m(\mathbf{e}; \alpha, \beta) &= 2\mathcal{R}_m(\mathbf{e} - m\delta_i; \alpha - 1, \beta) - \mathcal{R}_m(\mathbf{e} - 2m\delta_i; \alpha - 2, \beta) \\ &\quad - \mathcal{R}_m(\mathbf{e} - 2m\delta_i + \delta_{i-1}; \alpha, \beta - 1) + \mathcal{R}_m(\mathbf{e} - 2m\delta_i + \delta_{i+1}; \alpha, \beta - 1). \end{aligned} \quad (24)$$

The total exponent decreases by $m, 2m, 2m - 1, 2m - 1$, respectively. Thus (23)–(24), under any fixed reducible-coordinate choice, form a terminating induction certificate.

The same coefficient has the order-independent Laurent representation

$$\mathcal{R}_m(\mathbf{e}; \alpha, \beta) = \left[a^\alpha \varepsilon^\beta x_0^{-1} x_1^{-1} x_2^{-1} \right] \frac{x_0^{e_0} x_1^{e_1} x_2^{e_2}}{F_0 F_1 F_2}, \quad (25)$$

where

$$\frac{1}{F_i} = x_i^{-2m} \sum_{h \geq 0} \left(\frac{2ax_i^m - a^2 - \varepsilon x_{i-1} + \varepsilon x_{i+1}}{x_i^{2m}} \right)^h. \quad (26)$$

Unique normal form proves reduction-order independence. In the Laurent picture, interleavings of reductions at distinct coordinates are merely different orders for the same monomial; branch-count tuples, rather than interleavings, index the summands.

Expand

$$t_\varepsilon^{m+1} = \sum_{j=0}^{m+1} \binom{m+1}{j} \varepsilon^{2j} (q_0 q_1 q_2)^{m+1-j} (q_0 + q_1 + q_2)^j. \quad (27)$$

Put $\bar{q}_i = x_i^{m-1}(x_i^m - a)$. Extracting every scalar $2m$, define

$$\mathcal{C}_{m,j} := \left[a^\nu \varepsilon^{2m-2j} X_{\text{top}} \right] \text{NF} \left((\bar{q}_0 \bar{q}_1 \bar{q}_2)^{m+1-j} (\bar{q}_0 + \bar{q}_1 + \bar{q}_2)^j \right). \quad (28)$$

Then

$$D_m = \sum_{j=0}^{m+1} \binom{m+1}{j} (2m)^{3m+3-2j} \mathcal{C}_{m,j}. \quad (29)$$

Set $r = m - j$ for $0 \leq j \leq m$. Choose $\mathbf{s} = (s_0, s_1, s_2) \in \mathbb{N}^3$ with $|\mathbf{s}| = j$ and put $N_i = r + 1 + s_i$. Choosing the low term in exactly ℓ_i of the N_i factors of $\bar{q}_i = x_i^{2m-1} - ax_i^{m-1}$ gives

$$e_i = (2m - 1)N_i - m\ell_i, \quad 0 \leq \ell_i \leq N_i.$$

The terminating recurrence therefore gives

$$\begin{aligned} \mathcal{C}_{m,j} = \sum_{|\mathbf{s}|=j} \binom{j}{s_0, s_1, s_2} \sum_{0 \leq \ell_i \leq N_i} (-1)^{|\ell|} \prod_{i=0}^2 \binom{N_i}{\ell_i} \\ \cdot \mathcal{R}_m(\mathbf{e}; \nu - |\ell|, 2r). \end{aligned} \quad (30)$$

For $j = m + 1$, the requested ε -degree is negative, and $\mathcal{C}_{m,m+1} = 0$.

We now unfold (26). At source coordinate i , let A_i, B_i, U_i, V_i count, respectively, the four branch choices

$$2ax_i^m, \quad -a^2, \quad -\varepsilon x_{i-1}, \quad +\varepsilon x_{i+1},$$

and set

$$h_i = A_i + B_i + U_i + V_i, \quad T_i = U_{i+1} + V_{i-1}. \quad (31)$$

A tuple

$$(\mathbf{s}, \ell, \mathbf{A}, \mathbf{B}, \mathbf{U}, \mathbf{V}) \quad (32)$$

is admissible when all entries are nonnegative integers, $|\mathbf{s}| = j$, $0 \leq \ell_i \leq N_i$, and

$$|\ell| + \sum_i (A_i + 2B_i) = \nu, \quad (33)$$

$$\sum_i (U_i + V_i) = 2r, \quad (34)$$

$$e_i + mA_i + T_i - 2m(h_i + 1) = -1 \quad (i = 0, 1, 2). \quad (35)$$

Its signed multiplicity is

$$W = (-1)^{|\ell| + \sum_i (B_i + U_i)} 2^{\sum_i A_i} \binom{j}{s_0, s_1, s_2} \cdot \prod_{i=0}^2 \binom{N_i}{\ell_i} \frac{h_i!}{A_i! B_i! U_i! V_i!}. \quad (36)$$

Every expanded Laurent choice has exactly one branch-count tuple, and every admissible tuple has the requested parameter degrees and three Laurent exponents. Thus

$$\mathcal{C}_{m,j} = \sum_{\text{admissible tuples in (32)}} W. \quad (37)$$

The sum is finite by (33)–(34), and the description is exhaustive without double counting.

7.1 The local signed fiber identity

We use generalized binomial coefficients

$$\binom{z}{s} = \frac{z(z-1)\cdots(z-s+1)}{s!} \quad (s \geq 1), \quad \binom{z}{0} = 1,$$

and set $\binom{z}{s} = 0$ for $s < 0$. For a nonnegative integer upper argument, the usual out-of-range values are zero.

Lemma 7.1 (Local fiber identity). *For $N, n, \alpha \in \mathbb{N}$,*

$$\sum_{\substack{\ell, A, B \geq 0 \\ \ell + A + 2B = \alpha}} (-1)^{\ell+B} 2^A \binom{N}{\ell} \frac{(n+A+B)!}{n! A! B!} = (-1)^\alpha \binom{N-2n-2}{\alpha}. \quad (38)$$

Proof. Take the coefficient of z^α in

$$(1-z)^N \sum_{A, B \geq 0} \frac{(n+A+B)!}{n! A! B!} (2z)^A (-z^2)^B = (1-z)^N (1-2z+z^2)^{-n-1} \\ = (1-z)^{N-2n-2}.$$

The left coefficient is the sum in (38), and the right coefficient is $(-1)^\alpha \binom{N-2n-2}{\alpha}$. $\square$

Fix $(\mathbf{s}, \mathbf{U}, \mathbf{V})$ and put

$$n_i = U_i + V_i, \quad \lambda_i = N_i - 2n_i - 2, \quad \alpha_i = \frac{(2m-1)(N_i-1) + T_i}{m} - 2n_i. \quad (39)$$

Because $\sum_i (N_i - 1) = m + 2r$ and $\sum_i T_i = \sum_i n_i = 2r$,

$$\sum_i \alpha_i = \frac{(2m-1)(m+2r) + 2r}{m} - 4r = 2m-1 = \nu. \quad (40)$$

Equation (35) is precisely $\ell_i + A_i + 2B_i = \alpha_i$. Multiplying lemma 7.1 by the orientation factor $(-1)^{U_i} \binom{n_i}{U_i}$ turns its factorial into

$$\binom{n_i}{U_i} \frac{(n_i + A_i + B_i)!}{n_i! A_i! B_i!} = \frac{h_i!}{A_i! B_i! U_i! V_i!}.$$

Thus the local identity sums exactly over (ℓ_i, A_i, B_i) at fixed coupling orientations, including all signs and powers of two. Define

$$\mathcal{B}(\lambda, \alpha) = \begin{cases} (-1)^\alpha \binom{\lambda}{\alpha}, & \alpha \in \mathbb{N}, \\ 0, & \text{otherwise.} \end{cases} \quad (41)$$

7.2 Distinguished coordinate and transfer flow

The integrality condition in (39) is

$$T_i \equiv N_i - 1 = r + s_i \pmod{m}. \quad (42)$$

Write $T_i = r + s_i + mz_i$. Since $\sum_i T_i = 2r$ and $\sum_i (r + s_i) = m + 2r$,

$$z_0 + z_1 + z_2 = -1. \quad (43)$$

Also $0 \leq r + s_i \leq m$. Nonnegativity of T_i forces $z_i \geq -1$, and $z_i = -1$ is possible only if $s_i = j$ and $T_i = 0$.

Suppose first that $j \geq 1$. Two coordinates cannot both have $s_i = j$ because $\sum_i s_i = j$. Equation (43) therefore forces a unique distinguished coordinate d with

$$s_d = j, \quad T_d = 0, \quad s_i = 0, \quad T_i = r \quad (i \neq d). \quad (44)$$

By cyclic symmetry take $d = 0$, and write

$$u = U_0, \quad v = V_0, \quad k = n_0 = u + v.$$

The incoming-transfer equations give uniquely

$$U_1 = 0, \quad V_2 = 0, \quad V_1 = r - u, \quad U_2 = r - v, \quad n_1 = r - u, \quad n_2 = r - v. \quad (45)$$

Conversely, every nonnegative (k, u, v) satisfying the ranges below gives one transfer configuration.

At the distinguished coordinate,

$$(\lambda_0, \alpha_0) = (m - 1 - 2k, 2m - 1 - 2k),$$

whereas

$$(\lambda_1, \alpha_1) = (2u - r - 1, 2u), \quad (\lambda_2, \alpha_2) = (2v - r - 1, 2v).$$

Using

$$(-1)^\alpha \binom{\alpha - M}{\alpha} = \binom{M - 1}{\alpha} \quad (M \geq 1, \alpha \geq 0),$$

the local factors become

$$\mathcal{B}(\lambda_0, \alpha_0) = \binom{m - 1}{2(m - k) - 1}, \quad \mathcal{B}(\lambda_1, \alpha_1) = \binom{r}{2u}, \quad \mathcal{B}(\lambda_2, \alpha_2) = \binom{r}{2v}. \quad (46)$$

They are nonzero exactly when

$$\left\lceil \frac{m}{2} \right\rceil \leq k \leq m - 1, \quad 2u \leq r, \quad 2v \leq r. \quad (47)$$

The orientation multiplicity is $\binom{k}{u}$, and its sign is

$$(-1)^{U_0+U_1+U_2} = (-1)^{u+r-v} = (-1)^{r+k}.$$

Hence the exact signed fiber over (d, k, u, v) is

$$(-1)^{r+k} \binom{m-1}{2(m-k)-1} \binom{k}{u} \binom{r}{2u} \binom{r}{2v}. \quad (48)$$

The inequalities $2u, 2v \leq r$ and $u+v=k$ imply $k \leq r$. Summing (48) and then the three choices of d yields

$$\mathcal{C}_{m,j} = 3A_{m,r} \quad (1 \leq j \leq m). \quad (49)$$

When $j > \lfloor m/2 \rfloor$, one has $r = m - j < \lfloor m/2 \rfloor$, so the range is empty and

$$\mathcal{C}_{m,j} = 0. \quad (50)$$

The case $j = 0$ is not obtained by silently reusing the preceding classification. Here $r = m$, $s_i = 0$, and $T_i = m(1 + z_i)$. The nonnegative incoming-transfer patterns are exactly the cyclic permutations of

$$(0, m, m) \quad \text{and} \quad (0, 0, 2m). \quad (51)$$

For the first type, the unique zero coordinate plays the role of d , giving $3A_{m,m}$. For the second type, take $T_0 = 2m$, $T_1 = T_2 = 0$. The flow equations force $n_0 = 0$, and at coordinate zero

$$N_0 = m + 1, \quad \lambda_0 = m - 1, \quad \alpha_0 = 2m + 1.$$

Therefore

$$\mathcal{B}(\lambda_0, \alpha_0) = -\binom{m-1}{2m+1} = 0. \quad (52)$$

All three rotations of this exceptional type vanish, and hence $\mathcal{C}_{m,0} = 3A_{m,m}$.

Combining (29), (49)–(52), and $\mathcal{C}_{m,m+1} = 0$ proves

$$D_m = 3 \sum_{j=0}^{\lfloor m/2 \rfloor} \binom{m+1}{j} (2m)^{3m+3-2j} A_{m,m-j}, \quad (53)$$

which is (7). If $q = \lfloor m/2 \rfloor$, define

$$E_m = \sum_{j=0}^q \binom{m+1}{j} (2m)^{2(q-j)} A_{m,m-j}. \quad (54)$$

Then

$$D_m = 3(2m)^{3m+3-2q} E_m. \quad (55)$$

The derivation proves the certificate for every symbolic $m \geq 2$. It does not determine whether E_m , and therefore D_m , is nonzero in every degree.

8 The complete quartic fiber and its exact moment

Let p be a monic-centered quartic and $B_p = \mathbb{k}[x]/(p)$. At each distinct root α of p , the formal fixed trace eigenvalue is $p'(\alpha)$, repeated with local algebra length. Thus the formal fixed-point trace multiset is 0^4 exactly when every root of p is multiple. The only multiplicity partitions are 4 and $2 + 2$.

If $p = (x - \alpha)^4$, centering forces $-4\alpha = 0$, so $p = x^4$. If $p = (x - \alpha)^2(x - \beta)^2$, centering gives $2\alpha + 2\beta = 0$, hence $\beta = -\alpha$ and

$$p = (x^2 - \alpha^2)^2.$$

Writing $L = \alpha^2$ includes the first case as $L = 0$. The coefficient of x^2 is $-2L$, so L is unique. This proves the full-fiber classification in theorem 3.3.

8.1 Two source-level derivations of the quartic slope

As an internal source-level cross-check, specialize the all- m certificate:

$$H(2, 1) = 2, \quad A_{2,2} = -2, \quad A_{2,1} = 0,$$

which gives

$$D_2 = 3 \cdot 4^9(-2) = -1572864. \quad (56)$$

We now derive the same slope without the all-degree collapse.

Put

$$P_a(x) = (x^2 - a)^2, \quad g_a(x) = x(x^2 - a), \quad L_i = x_{i-1} - x_{i+1}.$$

Then $F_i = P_a(x_i) + \varepsilon L_i$, $q_i = 4g_a(x_i)$, and

$$t_\varepsilon = 4^3 g_0 g_1 g_2 + 4\varepsilon^2(g_0 + g_1 + g_2).$$

Consequently

$$t_\varepsilon^3 = \sum_{j=0}^3 \binom{3}{j} 4^{9-2j} \varepsilon^{2j} G_j, \quad G_j = (g_0 g_1 g_2)^{3-j} (g_0 + g_1 + g_2)^j. \quad (57)$$

Define

$$\mathcal{L}_j(a, \varepsilon) = \left[x_0^{-1} x_1^{-1} x_2^{-1} \right] \frac{G_j}{F_0 F_1 F_2}.$$

Then

$$\mathcal{C}_{2,j} = \left[a^3 \varepsilon^{4-2j} \right] \mathcal{L}_j(a, \varepsilon), \quad D_2 = \sum_{j=0}^3 \binom{3}{j} 4^{9-2j} \mathcal{C}_{2,j}. \quad (58)$$

Weights show that the selected coefficient is a constant multiple of a^3 , so set $a = 1$. With $P(x) = (x^2 - 1)^2$ and $g(x) = x(x^2 - 1)$, expand

$$\frac{1}{P_i + \varepsilon L_i} = \sum_{h_i \geq 0} (-1)^{h_i} \varepsilon^{h_i} \frac{L_i^{h_i}}{P_i^{h_i+1}}. \quad (59)$$

Introduce

$$\rho_{n,h}(d) = \left[x^{-1} \right] \frac{x^d g(x)^n}{P(x)^{h+1}} \quad (d \geq 0).$$

The identities

$$\frac{g^3}{P} = x^3(x^2 - 1), \quad \frac{g^3}{P^2} = \frac{x^3}{x^2 - 1}, \quad \frac{g^3}{P^3} = \frac{x^3}{(x^2 - 1)^3}$$

give

$$\begin{aligned}\rho_{3,0}(d) &= 0 & (d \geq 0), \\ \rho_{3,1}(0) &= \rho_{3,1}(2) = 1, & \rho_{3,1}(1) = 0, \\ \rho_{3,2}(0) &= \rho_{3,2}(1) = 0, & \rho_{3,2}(2) = 1.\end{aligned}$$

For $j = 0$, the total internal denominator order is $h_0 + h_1 + h_2 = 4$. The first vanishing forces every $h_i \geq 1$, leaving only the three cyclic arrangements of $(2, 1, 1)$. For $(h_0, h_1, h_2) = (2, 1, 1)$, the tensor coefficient is

$$(\rho_{3,2} \otimes \rho_{3,1} \otimes \rho_{3,1})(L_0^2 L_1 L_2).$$

Since

$$L_0 = x_2 - x_1, \quad L_1 = x_0 - x_2, \quad L_2 = x_1 - x_0,$$

we have $L_1 L_2 = -x_0^2 + x_0(x_1 + x_2) - x_1 x_2$. Extracting the x_0 -coefficient leaves $-(x_2 - x_1)^2$, and the remaining two coefficients give -2 . The other cyclic arrangements agree, and the total sign is $(-1)^4 = 1$. Hence

$$\mathcal{C}_{2,0} = -6. \tag{60}$$

For $j = 1$, each monomial of G_1 has coordinate powers g^3, g^2, g^2 , but the total internal denominator order is only two. A coordinate with $h_i = 0$ contributes either $g^2/P = x^2$ or $g^3/P = x^3(x^2 - 1)$, both polynomials, so no x_i^{-1} -term can occur. Nonvanishing would require all three $h_i \geq 1$, a contradiction. Thus $\mathcal{C}_{2,1} = 0$.

For $j = 2$, the internal ε -degree is zero. If

$$\mu_n = \left[x^{-1} \right] \frac{g(x)^n}{P(x)},$$

then $\mu_1 = 1, \mu_2 = \mu_3 = 0$. The square terms of G_2 have type $(3, 1, 1)$, and its cross terms have type $(2, 2, 1)$; their tensor coefficients vanish as $\mu_3 \mu_1^2$ and $\mu_2^2 \mu_1$. Hence $\mathcal{C}_{2,2} = 0$. The $j = 3$ summand already contains ε^6 and cannot contribute to ε^4 . Equation (58) now gives

$$D_2 = 4^9(-6) = -1572864 \tag{61}$$

without using the all-degree binomial collapse.

8.2 A direct normal-form ledger for the constant

Set $a = 0$ and $\varepsilon = 1$. Then

$$F_i = x_i^4 + x_{i-1} - x_{i+1}, \quad t = 64x_0^3 x_1^3 x_2^3 + 4(x_0^3 + x_1^3 + x_2^3).$$

Let $R(e_0, e_1, e_2)$ be the coefficient of $x_0^3 x_1^3 x_2^3$ in the normal form of $x_0^{e_0} x_1^{e_1} x_2^{e_2}$. Reduction in the three coordinates gives

$$\begin{aligned}R(e_0, e_1, e_2) &= R(e_0 - 4, e_1 + 1, e_2) - R(e_0 - 4, e_1, e_2 + 1), \\ R(e_0, e_1, e_2) &= -R(e_0 + 1, e_1 - 4, e_2) + R(e_0, e_1 - 4, e_2 + 1), \\ R(e_0, e_1, e_2) &= R(e_0 + 1, e_1, e_2 - 4) - R(e_0, e_1 + 1, e_2 - 4).\end{aligned} \tag{62}$$

The terminal value is $R(3, 3, 3) = 1$, and every other standard exponent triple has value zero. The complete ledger needed for t^3 is

Exponent pattern	R -value
(9, 9, 9)	-6
cyclic permutation of (9, 6, 6)	2
cyclic permutation of (9, 3, 3)	0
cyclic permutation of (6, 6, 3)	-1
permutation of (9, 0, 0)	0
permutation of (6, 3, 0)	0
(3, 3, 3)	1

For example,

$$R(9, 6, 6) = R(5, 7, 6) - R(5, 6, 7) = 1 - (-1) = 2,$$

and

$$R(9, 9, 9) = 2R(5, 10, 9) = 2(1 - 4) = -6.$$

The remaining rows require at most two reductions or have total degree nine without already being the top standard monomial.

Put $Q = 64x_0^3x_1^3x_2^3$ and $R_0 = 4(x_0^3 + x_1^3 + x_2^3)$. Grouping $(Q + R_0)^3$ by the preceding patterns gives

Source	Top-coefficient contribution
Q^3	$262144(-6) = -1572864$
$3Q^2R_0$	$3 \cdot 49152 \cdot 2 = 294912$
square terms in $3QR_0^2$	0
mixed terms in $3QR_0^2$	$3 \cdot 6144(-1) = -18432$
pure and 2 + 1 terms in R_0^3	0
fully mixed term in R_0^3	384

Therefore

$$C_2 = -1572864 + 294912 - 18432 + 384 = -1296000. \quad (63)$$

Combining (61) and (63) proves (8). Both computations are derivations inside the source proof: (56) is a specialization of the all-degree theorem, while (61)–(63) use the separate tensor-Laurent and normal-form routes.

9 Fixed moment, local multiplicity, and minimality

The next two lemmas have different conclusions and are used in that order.

Lemma 9.1 (Zero fixed moment). *On the fixed algebra (9), the fixed contribution to the formal period-three m -th trace moment is zero for every $m \geq 2$.*

Proof. All three cyclic coordinates agree on the fixed algebra, so

$$t_\varepsilon = q^3 + 3\varepsilon^2q.$$

Since $q^2 = 0$,

$$(q^3 + 3\varepsilon^2q)^m = 0 \quad (m \geq 2).$$

At $\varepsilon = 1$, this is the fixed contribution to the trace of $(Df^3)^m$. Hence the formal exact-period-three moment equals the raw f^3 -fixed moment $S_m(a, 1)$ after subtracting a zero fixed moment. $\square$

Lemma 9.2 (Local fixed multiplicity inside $\text{Fix}(f^3)$). *Let α be a root of p of multiplicity $r \geq 2$. The local length of $\text{Fix}(f^3)$ at the associated fixed point equals r , the fixed-scheme length at α .*

Proof. Set

$$\delta = x_0 - \alpha, \quad u = x_1 - x_0, \quad v = x_2 - x_0.$$

At $(\delta, u, v) = (0, 0, 0)$, the (u, v) -Jacobian of (F_1, F_2) is

$$\begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix},$$

which is invertible. Formal elimination therefore gives

$$u = -p(\alpha + \delta) + O(\delta^{2r-1}), \quad v = p(\alpha + \delta) + O(\delta^{2r-1}).$$

The remaining equation is

$$F_0 = p(\alpha + \delta) + v - u = 3p(\alpha + \delta) + O(\delta^{2r-1}). \quad (64)$$

Because $2r - 1 > r$ and the characteristic is zero, the order of (64) is exactly r . This is the local length of $\text{Fix}(f^3)$ at the fixed support. $\square$

Remark 9.3. Lemma 9.1 is a statement about a trace power, while lemma 9.2 is a statement about scheme length. The first does not imply the second. In particular, zero fixed moment is never used as a shortcut for subtracting reduced fixed support.

For the quartic fiber, the f^3 -fixed algebra has length $4^3 = 64$. By lemma 9.2, the fixed algebra contributes its full length 4, so the formal exact-period-three zero-cycle has length

$$64 - 4 = 60. \quad (65)$$

Every exact period-three orbit contains three points, and matrix trace is cyclically invariant along the orbit. Only after the pointwise exact-period subtraction do we divide by three:

$$\frac{-1296000 - 1572864L^3}{3} = -432000 - 524288L^3. \quad (66)$$

The normalized conjugacy calculation from section 4, with $m = 2$, gives

$$f_{2,L} \sim f_{2,M} \iff L^3 = M^3.$$

The coefficient of L^3 in (8) is nonzero. Hence the period-three moment separates precisely the normalized conjugacy classes on the complete quartic fiber. Periods one and two are constant there, so period three is minimal. This proves theorem 3.3.

10 Scope, open problem, and proof provenance

10.1 The open coefficient problem

The formula (53) reduces all-degree period-three recovery to a concrete combinatorial question:

$$D_m \stackrel{?}{\neq} 0 \quad \text{for every } m \geq 2. \quad (67)$$

No degree-independent sign, positivity, or nonvanishing proof is known in the present argument. Consequently theorem 3.2 proves an affine two-term law and an exact slope certificate, but it does not prove that period three separates the family in every degree.

10.2 Nonclaims

To make the theorem boundary explicit, this note does not claim any of the following:

- universal nonvanishing of D_m , or all-degree period-three separation;
- a theorem for all quartic Hénon maps, a global degree-four period cutoff, a global conjugacy classification, or global multiplier rigidity;
- discovery of the exceptional quartic family or of its period-one/two blindness;
- novelty of global residues, quotient traces, formal/dynatomic cycles, or low-period Hénon algebra;
- a historical-priority conclusion from a bounded no-hit search;
- an unstable- or saddle-multiplier theorem, an arithmetic height or finiteness result, a prime/zero statement, or a transfer-, Fredholm-, or Euler-product identity; or
- an all- m inference from any finite diagnostic or development calculation.

10.3 Proof and process provenance

Every theorem claim in this note is derived from the source-locked proof package with SHA-256

36f2edd5a1a25960a2c656adaeb07fbd4251c61b51c9e0a0382200460b589ba9.

That package received the independent source verdict `SOURCE_LOCK_PASS` in the review with SHA-256

5e66edcd7f33769f748c8b6bd6582aa7e05b3325329839c46bf3627945803d11.

No theorem in this note uses registered computational evidence.

The sole earlier registered audit was consumed and ended `REGISTERED_AUDIT_TERMINAL_FAIL`; no rerun is permitted, no raw result exists, and no result gate passed. Static forensic analysis attributes the failure with approximate confidence 0.98 to a deterministic Track-Q scalar-versus-pair endpoint defect, but the child stderr was not preserved. No scientific mismatch was recorded, and that absence is not evidence of agreement. This disclosure is non-evidentiary and supplies no support for any theorem above.

The historical positioning is equally bounded. Cantat–Dujardin already supply the quartic family and its period-one/two blindness; Friedland–Milnor supply the normalized Hénon framework; and multidimensional residues, quotient traces, and formal periodic cycles are prior methods. The contribution claimed here is limited to the source-proved period-three law and coefficient certificate and the scoped quartic minimal-separator theorem. The bounded literature search is not a proof of historical priority.

11 Conclusion

The exceptional family $f_{m,a}(x, y) = (y + (x^m - a)^2, x)$ is formally blind at periods one and two, yet its period-three trace residue has a rigid two-term dependence on the normalized coordinate a^{2m-1} . A complete Laurent fiber calculation turns the slope into the finite certificate (53). In degree four, the complete normalized fiber with formal fixed-point trace multiset 0^4 is $p = (x^2 - L)^2$, and its

exact affine period-three moment recovers L^3 . The separate local-multiplicity argument shows that the exact-period scheme has length 60, so the pointwise and cyclewise normalizations are both controlled.

The quartic theorem is complete within its stated fiber. The uniform theorem ends at a sharper frontier: proving or disproving $D_m \neq 0$ in every degree is an explicit open combinatorial problem.

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